Task title:

Mechanical Design of a Quadruped Robotic Prototype

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Introduction

This report outlines the mechanical design and development process for a quadruped robotic task. The objective is to translate locomotion principles into a functional mechanical system. By utilizing 3D modeling and structural design methodologies, this work establishes a foundational framework for robotic stability, kinematic control, and mechanical integration, demonstrating the practical application of basic robotics engineering.

Structural Design & Overview

The chassis is constructed as a rigid, rectangular frame to ensure optimal weight distribution and structural integrity. The design incorporates an articulated leg system, where each of the four limbs is engineered with two distinct pivot points—a hip joint and a knee joint. These joints are connected using precision fastening points, ensuring durability and facilitating the movements required for stable locomotion.

Degrees of Freedom (DOF)

The prototype features a total of 8 Degrees of Freedom (DOF), providing significant mechanical versatility:

-Hip Joint: Manages the rotational movement of the leg relative to the main chassis, enabling directional control.

-Knee Joint: Controls the extension and flexion of the lower leg, allowing for height adjustment and obstacle clearance.

-Mechanical Advantage: The 2-DOF-per-leg configuration allows for articulated, life-like movement, enabling the robot to perform more sophisticated maneuvers than a single-joint system.

Actuation and Torque

-Actuators: To support the 8-DOF structure, each joint is powered by an independent high-torque servo motor.

-Torque Calculation: The required torque (τ) for each joint is determined by the formula：

$$\tau = F \times d$$

(Where F represents the gravitational force acting on the robot’s mass, and d represents the lever arm distance.)

-The dual-joint distribution optimizes the workload, ensuring efficient energy consumption and operational stability during the stance phase.

Stability and Center of Gravity

The center of gravity is precisely positioned at the geometric center of the chassis. To maintain dynamic equilibrium, the gait control algorithm ensures that the projection of the center of gravity consistently remains within the "support polygon" created by the limbs currently in contact with the ground, preventing tipping during movement.

Proposed Gait Mechanism

The system utilizes an articulated walking gait. By synchronizing the "swing phase" (the elevation of a limb) with the "stance phase" (the weight-bearing motion), the robot achieves fluid and controlled movement. This approach mimics natural quadruped mechanics, offering a higher degree of stability and adaptability across different operational states compared to basic crawling methods.

Anticipated Mechanical Challenges

-Control Synchronization: Managing 8 independent motors requires precise algorithmic timing to ensure smooth motion and prevent mechanical binding.

-Structural Stress: The high-stress concentration at the joint interfaces necessitates the use of robust materials and precise fastening techniques to maintain long-term integrity.

-Dynamic Load Balancing: Maintaining stability during weight-shifting phases introduces potential vibrations, which must be mitigated through effective torque management and chassis rigidity.